Chapter 4
Multipoles, Electrostatics of Macroscopic Media, Dielectrics

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These notes follow Jackson's notation and organization: localized charge density $\rho(\mathbf{x}')$, electric dipole moment $\mathbf{p}$, traceless quadrupole tensor Q_{ij}, polarization $\mathbf{P}$, electric displacement $\mathbf{D}$, susceptibility χ_e, and permittivity ϵ. I kept the discussion close to the text, but tried to make the logical connections more explicit and the physical meaning a bit easier to see while still keeping the graduate-level flavor.

Prepared from the material spanning pp. 145–169 of Jackson, Chapter 4.

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What this chapter is really doing

Chapter 4 has two threads that at first look separate but are actually tightly connected. The first is the multipole expansion of localized charge distributions, which tells us how a finite source looks when observed from far away. The second is the macroscopic description of dielectrics, where many microscopic dipoles are averaged into continuum fields such as **P** and **D**. The bridge between the two viewpoints is the dipole moment: it is the first genuinely structural correction beyond the total charge, and in matter it becomes the natural macroscopic density of polarization.

1. Multipole Expansion

Consider a localized charge density $\rho(\mathbf{x}')$ that is nonzero only inside a finite region. For observation points outside that region, Jackson expands the potential in spherical harmonics as

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{4\pi}{2l+1} q_{lm} \frac{Y_{lm}(\theta,\phi)}{r^{l+1}}.$$

The coefficients are

$$q_{lm} = \int Y^*_{lm}(\theta',\phi')\, r'^l \rho(\mathbf{x}')\, d^3x'.$$

These are the spherical multipole moments. What the expansion says is simple: once one is far enough away, the detailed charge distribution no longer matters point by point. Instead, the source is encoded in a hierarchy of moments, each with a characteristic angular dependence and a characteristic falloff in r.

The first few moments make the physical meaning transparent. The $l = 0$ term is the monopole,

$$q_{00} = \frac{1}{\sqrt{4\pi}} \int \rho(\mathbf{x}')\, d^3x' = \frac{q}{\sqrt{4\pi}},$$

so the monopole coefficient is just the total charge. The $l = 1$ terms are the dipole moments and are linear combinations of the Cartesian dipole vector

$$\mathbf{p} = \int \mathbf{x}' \rho(\mathbf{x}')\, d^3x'.$$

The $l = 2$ terms are the quadrupole moments and can be expressed in terms of the traceless tensor

$$Q_{ij} = \int (3x'_i x'_j - r'^2 \delta_{ij}) \rho(\mathbf{x}')\, d^3x'.$$

Jackson's Cartesian form of the expansion is

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{r} + \frac{\mathbf{p}\cdot\mathbf{x}}{r^3} + \frac{1}{2} \sum_{i,j} Q_{ij} \frac{x_i x_j}{r^5} + \cdots \right].$$

This makes the hierarchy especially clear: the monopole falls as $1/r$, the dipole as $1/r^2$ in the potential, the quadrupole as $1/r^3$, and so on.

A nice point Jackson emphasizes is that the spherical and Cartesian multipoles are not identical objects arranged in different notation; they reflect different transformation properties. The Cartesian moments for fixed l are generally more numerous than the $2l+1$ spherical coefficients, and the spherical moments correspond to irreducible pieces under rotations. For $l = 2$, using a traceless quadrupole tensor is exactly what removes the redundant trace part and isolates the irreducible quadrupole content.

The electric field of a definite multipole term follows by $\mathbf{E} = -\nabla\Phi$. In spherical coordinates Jackson writes the components explicitly; for a dipole $\mathbf{p}$ aligned with the z axis one recovers the standard form

$$E_r = \frac{2p\cos\theta}{4\pi\epsilon_0 r^3}, \qquad E_\theta = \frac{p\sin\theta}{4\pi\epsilon_0 r^3}, \qquad E_\phi = 0.$$

In vector form, for a point dipole at $\mathbf{x}_0$,

$$\mathbf{E}(\mathbf{x}) = \frac{3\mathbf{n}(\mathbf{p}\cdot\mathbf{n}) - \mathbf{p}}{4\pi\epsilon_0|\mathbf{x}-\mathbf{x}_0|^3},$$

where $\mathbf{n}$ points from $\mathbf{x}_0$ to $\mathbf{x}$.

One subtle but important feature is origin dependence. A general charge distribution changes its higher multipole moments when the origin is shifted. Only the lowest nonvanishing moment is origin-independent. Thus, if the total charge is nonzero, the monopole is independent of origin but the dipole is not. If the total charge vanishes, then the dipole becomes origin-independent. This is why the dipole moment of a neutral system is especially meaningful.

Jackson then includes a very useful theorem about the volume average of the electric field over a sphere. If a sphere of radius R contains all the charge, then

$$\int_{r<R} \mathbf{E}(\mathbf{x})\, d^3x = -\frac{\mathbf{p}}{3\epsilon_0}.$$

If instead all the charge lies outside the sphere, then the average field over the spherical volume equals the field at the center:

$$\int_{r<R} \mathbf{E}(\mathbf{x})\, d^3x = \frac{4\pi}{3}R^3\mathbf{E}(0).$$

This result later becomes the key step in the Lorentz local-field argument for dielectrics. It is one of those places where what first looks like a technical detour turns out to be foundational for the rest of the chapter.

Finally, to make the dipole field consistent with the spherical average relation, Jackson writes the point-dipole field as

$$\mathbf{E}(\mathbf{x}) = \frac{1}{4\pi\epsilon_0}\left[\frac{3\mathbf{n}(\mathbf{p}\cdot\mathbf{n}) - \mathbf{p}}{|\mathbf{x}-\mathbf{x}_0|^3} - \frac{4\pi}{3}\mathbf{p}\,\delta(\mathbf{x}-\mathbf{x}_0)\right].$$

The delta-function term does not affect the field away from the dipole, but it fixes the correct integral properties at the source point. This is one of the earliest places where Jackson insists that singular idealizations must be handled distributionally if one wants the global statements to remain

correct.

2. Multipole Expansion of the Energy of a Charge Distribution in an External Field

If a localized charge distribution is placed in an external potential $\Phi(\mathbf{x})$, the electrostatic energy is

$$W = \int \rho(\mathbf{x})\Phi(\mathbf{x})\, d^3x.$$

Assuming the external potential varies slowly across the size of the source, one expands it about a chosen origin:

$$\Phi(\mathbf{x}) = \Phi(0) + \mathbf{x}\cdot\nabla\Phi(0) + \frac{1}{2}\sum_{i,j} x_i x_j \frac{\partial^2\Phi}{\partial x_i \partial x_j}(0) + \cdots .$$

Since $\mathbf{E} = -\nabla\Phi$ and the external field is source-free in the region of the distribution, Jackson rewrites the expansion so that the traceless quadrupole tensor appears naturally. The resulting energy expansion is

$$W = q\Phi(0) - \mathbf{p}\cdot\mathbf{E}(0) - \frac{1}{6}\sum_{i,j} Q_{ij}\frac{\partial E_j}{\partial x_i}(0) + \cdots .$$

This formula is conceptually beautiful because it shows how the external field probes successively finer structure in the source: the monopole couples to the potential itself, the dipole to the field, the quadrupole to the field gradient, and higher moments to higher spatial derivatives.

Jackson briefly remarks on nuclear physics here. The quadrupole moment of a nucleus measures departures from spherical symmetry, and because nuclear states can sit in electric field gradients, quadrupole couplings split otherwise degenerate levels. That remark is short in the text, but it helps make the abstract tensor formula feel concrete.

A second classic result follows immediately from the dipole term: the interaction energy of two point dipoles $\mathbf{p}_1$ and $\mathbf{p}_2$ separated by $\mathbf{x}_1 - \mathbf{x}_2$ is

$$W_{12} = \frac{\mathbf{p}_1\cdot\mathbf{p}_2 - 3(\mathbf{n}\cdot\mathbf{p}_1)(\mathbf{n}\cdot\mathbf{p}_2)}{4\pi\epsilon_0|\mathbf{x}_1 - \mathbf{x}_2|^3},$$

with $\mathbf{n}$ the unit vector along the line joining them. The sign depends strongly on orientation. Two parallel dipoles arranged head-to-tail attract; side-by-side they repel. So even before the chapter turns to media, one already sees how dipolar structure naturally produces anisotropic interactions.

3. Elementary Treatment of Electrostatics with Ponderable Media

Up to this point, electrostatics was discussed mostly for charges and conductors. In matter one must distinguish microscopic fields from macroscopic averaged fields. Jackson keeps the discussion intentionally elementary here, but the key macroscopic ideas already appear.

First, averaging does not spoil the electrostatic condition

$$\nabla \times \mathbf{E} = 0.$$

So the macroscopic electric field still derives from a scalar potential. The genuinely new ingredient is polarization. Under an applied field, molecules distort or reorient, producing microscopic dipole moments. The dipole moment per unit volume is the polarization,

$$\mathbf{P}(\mathbf{x}) = \sum_i N_i \langle \mathbf{p}_i \rangle.$$

If there is also free or excess charge, the macroscopic charge density is

$$\rho(\mathbf{x}) = \sum_i N_i \langle e_i \rangle + \rho_{\text{excess}}.$$

Usually the average molecular charge vanishes, so ρ denotes the free charge.

Jackson then constructs the potential of a polarized medium by treating each small volume element as carrying both charge density and dipole density. The exact macroscopic potential is

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \int d^3x' \left[\frac{\rho(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} + \mathbf{P}(\mathbf{x}') \cdot \nabla' \left(\frac{1}{|\mathbf{x} - \mathbf{x}'|} \right) \right].$$

After integrating by parts,

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{x}') - \nabla' \cdot \mathbf{P}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} \, d^3x'.$$

Thus the effective source density is not just ρ but

$$\rho_{\text{eff}} = \rho - \nabla \cdot \mathbf{P}.$$

This naturally leads to the macroscopic Gauss law

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0}(\rho - \nabla \cdot \mathbf{P}).$$

Introducing the electric displacement,

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P},$$

one gets the standard macroscopic equation

$$\nabla \cdot \mathbf{D} = \rho.$$

So $\mathbf{D}$ is designed precisely so that only free charge appears on the right-hand side.

For a linear, isotropic medium Jackson assumes

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E},$$

where χ_e is the electric susceptibility. Then

$$\mathbf{D} = \epsilon \mathbf{E}, \qquad \epsilon = \epsilon_0(1 + \chi_e).$$

The ratio $\epsilon/\epsilon_0 = 1 + \chi_e$ is the dielectric constant or relative permittivity. In a uniform medium, the field equation becomes

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon}.$$

So within a homogeneous dielectric all the vacuum formulas reappear after replacing ϵ_0 by ϵ.

The real new difficulty comes at interfaces. Jackson states the boundary conditions in the now-familiar form

$$(\mathbf{D}_2 - \mathbf{D}_1) \cdot \mathbf{n}_{21} = \sigma, \qquad (\mathbf{E}_2 - \mathbf{E}_1) \times \mathbf{n}_{21} = 0,$$

where σ is the free surface charge. These are exactly the macroscopic boundary conditions one expects from Maxwell's equations, but here they are deployed in electrostatic problems with matter.

4. Boundary-Value Problems with Dielectrics

Jackson then works through two representative problems. The first is a point charge q in a dielectric ϵ_1 at distance d from a planar interface with a second dielectric ϵ_2. Using a method of images adapted to dielectrics, he writes for the region containing the charge

$$\Phi = \frac{1}{4\pi\epsilon_1}\left(\frac{q}{R_1} + \frac{q'}{R_2}\right), \qquad z > 0,$$

and for the other region

$$\Phi = \frac{1}{4\pi\epsilon_2}\frac{q''}{R_1}, \qquad z < 0.$$

Enforcing the continuity of tangential E and normal D at the interface gives

$$q' = -\left(\frac{\epsilon_2 - \epsilon_1}{\epsilon_2 + \epsilon_1}\right) q, \qquad q'' = \left(\frac{2\epsilon_2}{\epsilon_2 + \epsilon_1}\right) q.$$

This is a very clean generalization of the conducting-plane image method. In the limit $\epsilon_2 \gg \epsilon_1$, the second medium behaves almost like a conductor and the image charge approaches the familiar conducting result.

Jackson then interprets the interface physically in terms of polarization charge. Since the susceptibility jumps across the interface, there is a surface polarization charge

$$\sigma_{\text{pol}} = -(\mathbf{P}_2 - \mathbf{P}_1) \cdot \mathbf{n}_{21}.$$

For this geometry he obtains an explicit surface charge density proportional to

$$\sigma_{\text{pol}} \propto \frac{d}{(\rho^2 + d^2)^{3/2}}.$$

So the image construction is not just a mathematical trick; it is representing the actual induced polarization of the second medium.

The second example is a dielectric sphere of radius a and permittivity ϵ placed in a uniform external field $E_0\hat{z}$. Because the problem is axially symmetric, the potential is expanded in Legendre polynomials inside and outside. Matching tangential E and normal D at $r = a$ leaves only the $l = 1$ term. Jackson finds

$$\Phi_{\text{in}} = -\left(\frac{3}{\epsilon/\epsilon_0 + 2}\right) E_0 r \cos\theta,$$

$$\Phi_{\text{out}} = -E_0 r \cos\theta + \left(\frac{\epsilon/\epsilon_0 - 1}{\epsilon/\epsilon_0 + 2}\right) E_0 \frac{a^3}{r^2} \cos\theta.$$

Thus the internal field is uniform,

$$E_{\text{in}} = \frac{3}{\epsilon/\epsilon_0 + 2} E_0,$$

and the outside field is the applied field plus that of a dipole at the origin with moment

$$\mathbf{p} = 4\pi\epsilon_0 \left(\frac{\epsilon/\epsilon_0 - 1}{\epsilon/\epsilon_0 + 2}\right) a^3 E_0\, \hat{z}.$$

The sphere therefore behaves like a polarizable object whose induced dipole is proportional to the applied field.

Jackson immediately uses the same algebra to discuss a spherical cavity in a dielectric. Replacing $\epsilon/\epsilon_0 \to \epsilon_0/\epsilon$ gives a field inside the cavity

$$E_{\text{in}} = \frac{3\epsilon}{2\epsilon + \epsilon_0} E_0,$$

which is larger than E_0 if $\epsilon > \epsilon_0$. That contrast between the dielectric sphere and the cavity is physically instructive: matter inserted into a field tends to reduce the field inside itself, while a cavity carved inside a polarized medium experiences an enhanced local field.

5. Molecular Polarizability and Electric Susceptibility

Now the chapter turns microscopic again. The constitutive law $\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$ is macroscopic, but what determines χ_e? Jackson addresses this by introducing the molecular polarizability γ_{mol}.

A key distinction is between the macroscopic field $\mathbf{E}$ and the actual field acting on a given molecule. The local field is written as

$$\mathbf{E}_i = \mathbf{E}_{\text{near}} - \mathbf{E}_P,$$

where the second term is the contribution of the nearby molecules replaced by a smooth polarization continuum. Using the spherical-average result from Section 4.1, Jackson finds for a small spherical cavity in a uniformly polarized medium

$$\mathbf{E}_P = -\frac{\mathbf{P}}{3\epsilon_0}.$$

Hence

$$\mathbf{E}_i = \frac{1}{3\epsilon_0}\mathbf{P} + \mathbf{E}_{\text{near}}.$$

Lorentz showed that for a simple cubic lattice the near-field term vanishes, and Jackson treats $\mathbf{E}_{\text{near}} \approx 0$ as a good working approximation for many materials. Then the local field is

$$\mathbf{E}_i \approx \mathbf{E} + \frac{\mathbf{P}}{3\epsilon_0}.$$

If the average molecular dipole is proportional to the local field,

$$\langle \mathbf{p}_{\text{mol}} \rangle = \epsilon_0 \gamma_{\text{mol}} (\mathbf{E} + \mathbf{E}_i),$$

and since

$$\mathbf{P} = N \langle \mathbf{p}_{\text{mol}} \rangle,$$

one arrives at

$$\mathbf{P} = N\gamma_{\text{mol}} \left(\epsilon_0 \mathbf{E} + \frac{1}{3}\mathbf{P} \right).$$

Solving for χ_e yields

$$\chi_e = \frac{N\gamma_{\text{mol}}}{1 - \frac{1}{3}N\gamma_{\text{mol}}}.$$

Equivalently,

$$\gamma_{\text{mol}} = \frac{3}{N} \left(\frac{\epsilon/\epsilon_0 - 1}{\epsilon/\epsilon_0 + 2} \right).$$

This is the Clausius–Mossotti relation. It provides the bridge between the microscopic parameter γ_{mol} and the macroscopic dielectric constant. Jackson remarks that it works best for dilute systems such as gases and only approximately for dense liquids and solids.

6. Models for the Molecular Polarizability

Jackson then sketches two classical models.

The first is induced polarization. A charge e harmonically bound with natural frequency ω_0 obeys

$$F = -m\omega_0^2 x, \qquad m\omega_0^2 x = eE.$$

Hence the induced dipole moment is

$$\mathbf{p}_{\text{mol}} = ex = \frac{e^2}{m\omega_0^2}\mathbf{E}.$$

Thus the molecular polarizability is

$$\gamma_{\text{mol}} = \frac{e^2}{m\omega_0^2 \epsilon_0},$$

and for many charges in a molecule,

$$\gamma_{\text{mol}} = \frac{1}{\epsilon_0} \sum_j \frac{e_j^2}{m_j \omega_j^2}.$$

This explains why molecular polarizability has the dimensions of volume and why its magnitude is usually comparable to molecular size.

Jackson also checks the role of thermal motion. For a harmonically bound charge in a classical Boltzmann distribution, the average dipole moment still comes out exactly

$$\langle \mathbf{p}_{\text{mol}} \rangle = \frac{e^2}{m\omega_0^2} \mathbf{E},$$

so the induced polarizability is independent of temperature within this model.

The second mechanism is orientation polarization. Molecules with permanent dipole moment $\mathbf{p}_0$ tend to align with an applied field but are randomized by thermal agitation. Using the Boltzmann factor with Hamiltonian

$$H = H_0 - \mathbf{p}_0 \cdot \mathbf{E},$$

Jackson derives, for weak fields,

$$\langle \mathbf{p}_{\text{mol}} \rangle \approx \frac{1}{3} \frac{p_0^2}{kT} \mathbf{E}.$$

So orientation polarizability varies as $1/T$. Combining both mechanisms,

$$\gamma_{\text{mol}} \approx \gamma_i + \frac{1}{3\epsilon_0} \frac{p_0^2}{kT}.$$

This is the basic reason why polar substances show a temperature-dependent molecular polarizability while nonpolar substances do not. Jackson's Figure 4.10 illustrates exactly that: one obtains approximately a straight line when plotting γ_{mol} against $1/T$ for polar molecules.

7. Electrostatic Energy in Dielectric Media

The last section revisits energy, now in matter. The vacuum formula

$$W = \frac{1}{2} \int \rho(\mathbf{x}) \Phi(\mathbf{x}) \, d^3x$$

cannot simply be taken over unchanged because work is required not only to assemble free charge but also to polarize the medium itself.

Jackson starts from the differential work associated with a small change in free charge density,

$$\delta W = \int \delta\rho(\mathbf{x}) \Phi(\mathbf{x}) \, d^3x.$$

Since $\nabla \cdot \mathbf{D} = \rho$, one has $\delta\rho = \nabla \cdot (\delta\mathbf{D})$, and after integration by parts,

$$\delta W = \int \mathbf{E} \cdot \delta\mathbf{D}\, d^3x.$$

Therefore the total electrostatic energy can be written formally as

$$W = \int d^3x \int_0^{\mathbf{D}} \mathbf{E} \cdot \delta\mathbf{D}.$$

If the medium is linear, then

$$\mathbf{E} \cdot \delta\mathbf{D} = \frac{1}{2}\delta(\mathbf{E} \cdot \mathbf{D}),$$

so

$$W = \frac{1}{2}\int \mathbf{E} \cdot \mathbf{D}\, d^3x.$$

Thus the familiar quadratic energy density survives only in linear media. Jackson is careful here: for nonlinear media the energy depends on the path in constitutive space, and one must use the integral form rather than the simple quadratic expression.

He then considers the change in energy when a linear dielectric body is inserted into a field whose sources are fixed. Let $\mathbf{E}_0, \mathbf{D}_0$ be the field before insertion and $\mathbf{E}, \mathbf{D}$ after. After a neat integration-by-parts argument using the fact that the free source charge is unchanged, Jackson finds

$$W = \frac{1}{2}\int (\mathbf{E} \cdot \mathbf{D}_0 - \mathbf{D} \cdot \mathbf{E}_0)\, d^3x.$$

Restricting to the volume V_1 of the dielectric body and using $\mathbf{D} = \epsilon\mathbf{E}$ there, this becomes

$$W = -\frac{1}{2}\int_{V_1} (\epsilon_1 - \epsilon_0)\mathbf{E} \cdot \mathbf{E}_0\, d^3x.$$

If the surrounding medium is free space, Jackson rewrites it with the polarization as

$$W = -\frac{1}{2}\int_{V_1} \mathbf{P} \cdot \mathbf{E}_0\, d^3x.$$

So the energy density of a polarizable dielectric in an external field with fixed sources is

$$w = -\frac{1}{2}\mathbf{P} \cdot \mathbf{E}_0.$$

The factor 1/2 is crucial. A permanent dipole in a fixed field has energy $-\mathbf{p} \cdot \mathbf{E}$, but an induced dipole builds up gradually with the applied field, so the work accumulates with the extra factor of one half.

Jackson then turns to forces. If the free charges are fixed, a dielectric with $\epsilon_1 > \epsilon_0$ is drawn into regions of stronger field. The force on a generalized displacement ξ is

$$F_\xi = -\left(\frac{\partial W}{\partial \xi}\right)_Q.$$

But if instead the electrodes are held at fixed potential by a battery, external work is supplied as the dielectric moves and charge flows. Repeating the bookkeeping leads to the opposite sign,

$$F_\xi = +\left(\frac{\partial W}{\partial \xi}\right)_V .$$

Jackson summarizes the relation symbolically as

$$\delta W_V = -\delta W_Q.$$

This distinction between fixed-charge and fixed-potential conditions is one of the most important conceptual points in practical electrostatics. The same physical motion can be described by two different energy functions, depending on what the external circuit is holding fixed.

Closing synthesis

This chapter begins with the far-field expansion of a localized source and ends with the macroscopic energy of polarizable bodies. The common thread is that dipoles sit at the center of both stories. In the first half, the dipole is the first structural correction beyond total charge and often the first origin-independent moment of a neutral system. In the second half, polarization is just dipole moment density, and the constitutive law of a dielectric is therefore a continuum version of the dipole response of molecules.

The chapter also quietly assembles several of the standard formulas of electromagnetism into one chain: the spherical average of a dipole field, the Lorentz local field, the Clausius–Mossotti relation, the dielectric sphere in a uniform field, and the energy formulas for dielectric bodies. Once those are all in place, the later theory of macroscopic electrodynamics feels much less mysterious.